Lecture 1

Introduction to concepts of probability and statistics

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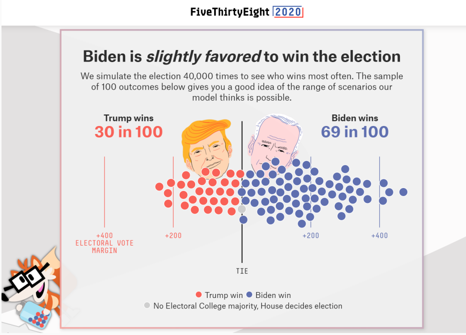
LPO.8800: Statistical Methods for Education Research

Why study statistics?

Why study statistics?

- Data, statistics, and visualizations of data are *everywhere*—and increasingly so.
- Technology has vastly increased the availability of raw data and raised the sophistication of everyday statistical reporting.
- AI can do a lot for you, but...

Why study statistics? (2020)



Why study statistics? (2016)

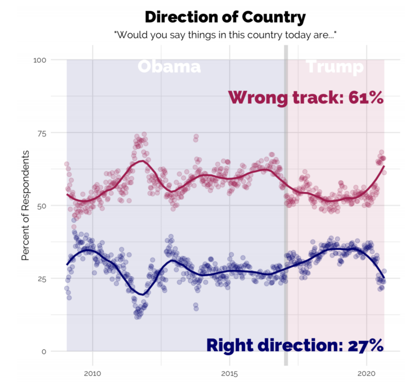
Who's ahead in each state and by how much

Our win probabilities come from simulating the election 10,000 times, which produces a distribution of possible outcomes for each state. Here are the expected margins of victory. The closer the dot is to the center line, the tighter the race. And the wider the bar, the less certain the model is about the outcome.

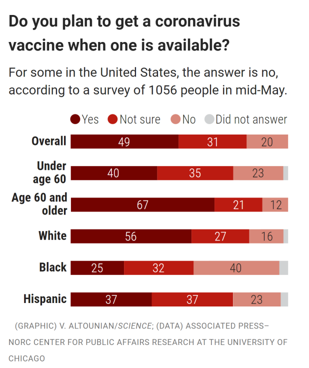
KEY AVERAGE — 80% CHANCE OUTCOME FALLS IN THIS RANGE

	Expected margin of victory	Chance of tipping election
Florida	D+1.4	16.4%
Ohio	D+1.4	12.0%
Pennsylvania	D+4.8	10.2%
North Carolina	D+0.6	8.1%
Michigan	D+5.8	7.6%
Wisconsin	D+5.3	6.4%
Virginia	D+6.1	6.1%
Colorado	D+5.3	3.7%
Minnesota	D+7.1	3.3%
New Jersey	D+9.5	3.2%
Arizona	R+2.8	2.7%
Iowa	D+0.3	2.5%
New Hampshire	D+4.7	2.5%
Georgia	R+4.0	2.4%

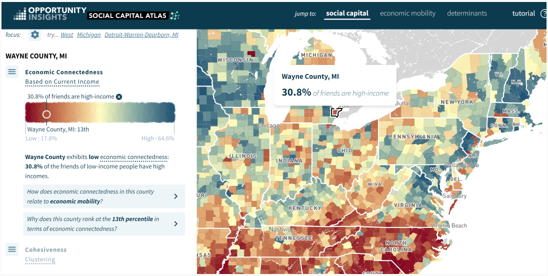
Why study statistics?



Why study statistics?

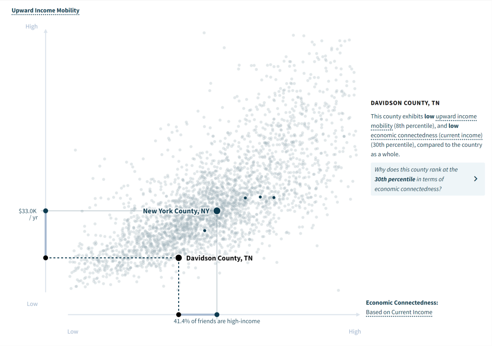


Why study statistics?



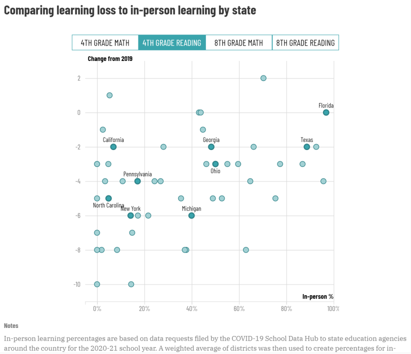
Source: Social Capital Atlas `https://socialcapital.org/`, Chetty et al. (2022). Uses data on 21 billion friendships from Facebook to measure social connectedness across income levels.

Why study statistics?



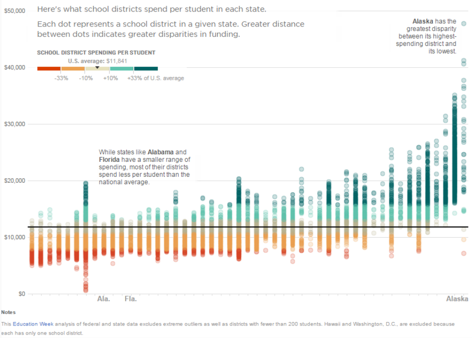
Source: Social Capital Atlas `https://socialcapital.org/`

Why study statistics?

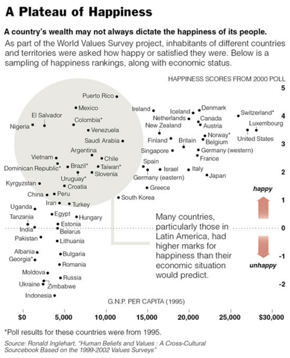


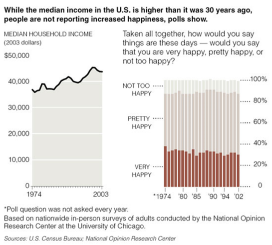
Source: Chalkbeat

Why study statistics?



Why study statistics?





Why study statistics?

- This is especially true in education, as student- and school-level data, accountability, and public reporting have all grown in use (NCLB).
- The use (and misuse) of data is ubiquitous in education policy.
- A strong foundation in statistics will help you to be an intelligent producer and consumer of data analysis.

Descriptive vs. inferential statistics

Descriptive vs. inferential statistics

- Statistics deals primarily with *variation* and/or *uncertainty* in an *outcome, characteristic*, or phenomenon of interest.
- A collection of observations on these outcomes is referred to as **data** or a **data set**.
- The **unit of observation** in one's data set may depend on the research question of interest (and/or data availability).

Descriptive vs. inferential statistics

- Statistical methods can be classified as **descriptive** or **inferential**.
- **Descriptive statistics** are used to *describe* outcomes in a population or sample (e.g., central tendency, variation, distribution shape; overall or by subgroup; correlation).
- **Inferential statistics** are used to make *inferences* or *predictions* about a population larger than that observed in the data.

Descriptive vs. inferential statistics

- **Population**: the universe of outcomes of interest
  - Commuting time for Vanderbilt graduate students
  - Taxable incomes of U.S. households
  - Math ability of 4th grade students
  - Favored candidate of registered voters in a political race
  - Self-rated health of surgical patients
- Notice each of these examples includes an *outcome* and a *unit of observation* (and often a time/place). Typically specifying only one of these is not sufficient.
- The researcher may or may not observe (or be able to observe) the population of interest.

Descriptive vs. inferential statistics

- **Sample**: a subset of the population, chosen at random or by some other method.
- Descriptive statistics can be conducted on either a population or a sample.
- The key difference is how these statistics are used / interpreted.

Descriptive vs. inferential statistics

- It may be impossible, or cost-prohibitive, to observe the full population. In these cases a sample can be used to make inferences about the population.
- An important step in inferential statistics is the *quantification of uncertainty* (e.g., standard errors, “margin of error”, confidence intervals). Covered in Lecture 5 and following.

Be precise when defining populations

In data analysis it is important to be precise in defining:

- The unit of observation (level at which the data are observed)
- The outcome of interest (and how it is measured)
- The population of interest (who, what, when, where)
- How the sample is drawn (when applicable)

Be precise when defining populations

Box 9. An example of the Complexity of Describing Constructs

How many eighth grade English teachers are in your schools?

Component	Issues to be clarified
How many	Does "how many" refer to a head count or full-time equivalent (FTE) count?
Eighth grade	Does "eighth grade" include classes with seventh-, eighth-, and ninth-grade students; or just classes with only eighth graders?
English	Does "English" include reading and writing classes? Special education English language classes? Other language arts classes? English as a Second Language classes?
Teachers	Do "teachers" include only certified teachers? Only certified English teachers? Certified teaching assistants? Only teachers assigned to teach classes/students this grading period?
Are	At what point in time should the answer be valid? At the beginning or end of the current or previous school year?
In	Does this include teachers of students cross-enrolled in virtual settings? What if someone teaches English in more than one school—does he or she get counted more than once?
Your	Does this mean only schools under the authority of the state or local education agency, or does it include all schools within the boundaries of the state or locality?
Schools	Are special education schools included? Correctional institutions that grant educational degrees? Other residential facilities? Cross-enrolled virtual settings? Private schools?

Test yourself

Descriptive or inferential? Unit of observation? Population of interest?

- What is the average annual household income in this class?
- What share of adults in the United States have employer-provided health insurance?
- What fraction of the incoming freshman class in NYC public schools in 2004 graduated in four years?
- What is the 4-year high school graduation rate in the United States?

Descriptive vs. inferential statistics

In this course:

- Lectures 1 - 3 and some of 9-11: descriptive statistics
- Lectures 4 - 8 and some of 9-11: inferential statistics

Variables and measurement

Variables and measurement

The “outcomes” described above are also called **variables** (or *random variables*) because they can vary from one unit of observation to another.

- Many variables are inherently numeric; others are not.
- **Categorical** variables can be assigned numeric values for convenience.
- Examples: male / female, employed / unemployed, strongly disagree … strongly agree, marital status

Variables and measurement

The possible values a variable can take on (and their meaning) form its **measurement scale**.

- The measurement scale is important, as it dictates which kinds of statistical analysis are appropriate and their interpretation.
- Scales can be characterized in a number of ways.
  - Quantitative, categorical, or qualitative
  - Nominal, ordinal, interval or ratio
  - Discrete (“count”) or continuous

Variables and measurement

One dimension is whether the variable is quantitative, categorical, or qualitative:

- A **quantitative** variable is on a numeric scale, where the numeric values express the magnitude of some property or characteristic.
- A **categorical** variable consists of a number of distinct categories that may or may not have a natural ordering. It is usually feasible to assign numeric values to categories.
- A **qualitative** variable lacks natural ordering and it is difficult or impossible to assign numeric values to them (e.g., text data).

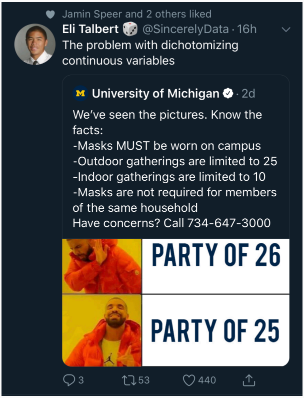
Note some texts will call categorical variables "qualitative."

Nominal scales

Categorical variables that lack a natural ordering are on a **nominal scale**.

- Categories simply recognize *differences*
- Numeric values can be assigned to categories; the values themselves are arbitrary
- Order is not meaningful; values cannot be compared
- A **dichotomous** variable has *two* categories, 0-1 (a special case). In practice called a **dummy, binary** or **indicator** variable.
  - It is often useful to create a series of binary variables from a categorical variable. E.g. "religion" could be measured using dummies ("Are you Christian?" "Are you Jewish?" "Are you Muslim?" etc).

Dichotomous danger



Nominal scales

Variables with a **nominal scale**:

- Gender identity 1 = Woman, 2 = Man, 3 = Non-binary, 4 = Transgender, 5 = Another term
- State of residence 1 = NY, 2 = NJ, 3 = CT
- High school graduate 1 = Yes, 2 = No
- Employment status: 0 = unemployed, 1 = employed
- Favorite type of music: 1 = Rock, 2 = Classical, 3 = Jazz, 4 = Country, 5 = Other
- Type of health insurance (if any): Medicare, Medicaid, CHIP, military, state-sponsored, private insurance, etc.

Ordinal scales

Categorical variables with a natural ordering are on an **ordinal scale**.

- Numeric values are assigned to categories, *and* their order is meaningful.
- The values themselves are *arbitrary*, but higher value implies *more* of some property.
- Numeric values can be compared, but only to determine which has more or less of some property.
- The *interval* between numeric values is not meaningful, and thus their difference and ratio are not meaningful.

Ordinal scales

Variables with an **ordinal scale**:

- NAEP math skills 1 = basic, 2 = proficient, 3 = advanced
- Job satisfaction 1 = very dissatisfied, 2 = somewhat dissatisfied, 3 = neutral, 4 = somewhat satisfied, 5 = very satisfied (a **Likert**-type item)
- Self-rated health 1 = poor, 2 = fair, 3 = good, 4 = very good, 5 = excellent
- Level of education completed 1 = less than HS, 2 = HS, 3 = some college, 4 = college or higher

Ordinal scales

In education, state assessments commonly use an ordinal scale to indicate performance levels. Ex:

- New York State: Levels 1 to 4 (3 or 4 is considered “proficent”)
- Tennessee: Level 1 = below grade level, 2 = approaching, 3 = on-track, 4 = mastered

More on this later.

Interval scales

Quantitative variables may have an **interval** or **ratio** scale:

- Numeric values have meaning, and can be used to compare magnitude of some property
- The intervals between numeric values are meaningful: equal increments on the scale implies equal intervals of some property
- The *ratio* of two values *may or may not* be meaningful (if they are, the measure can also be said to have a **ratio** scale)
- Most quantitative variables have a ratio scale, but some are interval but not ratio.

Interval scales

Variables with an **interval** (and **ratio**) scale:

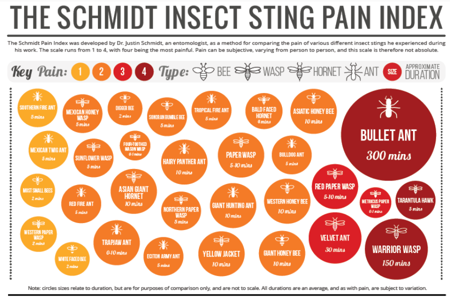
- days of instruction (150 - 205)
- annual income (\$0 to ∞?)
- calories consumed in a day (e.g., 1000 - 10000)
- Can make ratio comparisons: Person A consumed twice as many calories (3000) as Person B (1500)
- Ratio scales have a clear "zero value"

Interval scales

Some scales are **interval** but not ratio:

- Temperature is the most common example:
  - 60°F is 20°F more than 40°F
  - 80°F is 20°F more than 60°F
  - But 80°F is not "four times as warm" as 20°F.
  - Arbitrary zero point: "temperature" does not begin at a specific floor (e.g. zero) that would enable ratio comparisons.
- Date, measured from an arbitrary starting point
- Opinion and attitude scales (maybe, if not just ordinal)
- IQ, math achievement (maybe, if not just ordinal)

(Not so) fun scales



Discrete vs. continuous

Quantitative variables can be discrete ("count") or continuous:

- **discrete**: the variable can take on a *countable* number of values (e.g., values can be represented by integers). Does not take on negative or fractional values.
- **continuous**: the variable can take on a *continuum* of values (e.g., all values between some a and b)

Discrete vs. continuous

It is useful to separate in your mind features of the *underlying property* you are measuring and the scale of the *measurement tool* being used to measure it (and/or the observed data). Ex: height

- The underlying property being measured is continuous
- The scale may be discrete—e.g. rounded to the nearest inch
- The large number of possible values in the data lead us to treat the measure as continuous when analyzing

Confusing, I know...

Test yourself

Quantitative, categorical, or qualitative? Nominal, ordinal, interval, ratio? Discrete or continuous?

- Height
- Profession
- Percentage of HS seniors who have never smoked cigarettes
- Annual rainfall in Savannah, GA to the nearest inch
- Number of points scored in a football game
- Narcissism inventory scale
- Verbal aptitude as measured by SAT verbal
- Out-of-pocket costs in the past year for medical expenses
- Age
- Eye color

Some advice

With quantitative measures, always be clear about the unit of measurement (dollars, acres, pounds, 1000s of dollars, etc.)

Preserve the original level of detail in quantitative measures, even if you create cruder versions of them (e.g., rounded or ordinal measures). It's easy to go from a fine measure to something less fine, but hard to go the other direction.

- ex: if individuals' heights are known (6'1", 5'3", etc.—a ratio scale) one throws information away by simply classifying individuals as "short" or "tall" (an ordinal scale)
- Preserving the original variable in your data will permit you to create other versions of the variable on a different scale

From categorical to quantitative measures

Categorical measures can become quantitative in some circumstances:

- Creating a set dummy variables from a categorical variable (e.g., religion)
- Midpoint approximation of an ordinal variable (e.g., income categories)
- Multi-item scales
- Aggregation: variables that are categorical at the individual level may become quantitative when aggregated (e.g., type of health coverage)

Perils of proficiency

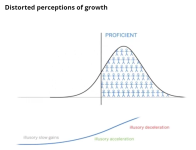
In educational testing, students are often assigned an ordinal performance level (e.g., 1-4) based on their scale score. A threshold is set for proficiency (e.g., 3+ is proficient).

Policymakers often aggregate these at the school, district, or state level and report the "percent proficient." Andrew Ho (Harvard GSE) has written extensively about the perils of using proficiency when comparing results across locations or over time: `https://www.gse.harvard.edu/ideas/usable-knowledge/16/01/when-proficient-isnt-good`

Perils of proficiency

Problems:

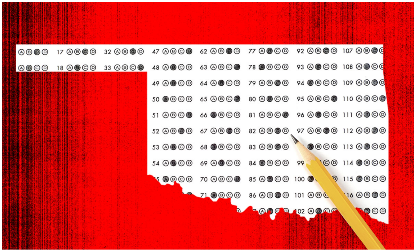
- Proficiency thresholds are arbitrary and may change.
- Comparing *growth* over time is complicated: schools/districts with $\sim$ 50% proficiency will see more growth for a given change in performance than will those with low/high proficiency rates.
- Comparing changes in *achievement gaps* over time is complicated for the same reason.



Perils of proficiency

Skyrocketing Test Gains in Oklahoma Are Largely Fiction, Experts Say

Frustrated local officials say state chief Ryan Walters has failed to explain how the state lowered proficiency targets.



Eamonn Fitzmaurice/The 74

By Linda Jacobson | August 21, 2024

Perils of proficiency

By Linda Jacobson | August 21, 2024

Updated August 23

Oklahoma school districts got some shocking, but welcome news this month when the state released results of student tests from last school year.

Student performance, especially in English language arts, appeared to have skyrocketed. A highlight: An impressive 51% of third graders scored proficient or better, compared to 29% last year. The reported jump came a full eight years before the majority of Oklahoma students are expected to reach proficiency under the state's plan to meet federal accountability laws.

But elation quickly turned to disbelief as local officials took a closer look at the data.

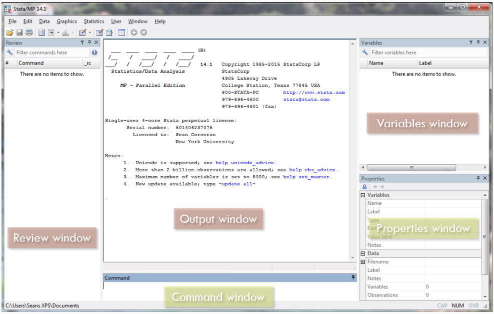
"Nobody makes jumps of that size," said an assessment director from a school system near Oklahoma City. The official asked not to be named because she does not want to "put a target" on her district.

To put the outsized gains in perspective, The 74 asked Andrew Ho, a leading testing expert at Harvard University, to review the results.

Data analysis in Stata

A crash course in Stata

You can view my introduction to Stata on Youtube:
`https://youtu.be/160nBMavcHE`



A crash course in Stata

See also the accompanying handout on Github *Useful Stata commands*. These cover:

- Stata interface
- Syntax
- Working with .dta and .do files
- Reviewing contents of a dataset
- Simple descriptive statistics
- Dropping and keeping variables
- Using log files
- Creating new variables
- Variable manipulation
- ...and more

Introduction to NELS-88

A simple dataset we will use throughout the course is an extract from the National Education Longitudinal Study of 1988 (NELS-88) published by the National Center for Education Statistics.
`http://nces.ed.gov/surveys/nels88/`

National Education Longitudinal Study of 1988

NELS 88 Overview

A nationally representative sample of eighth-graders were first surveyed in the spring of 1988. A sample of these respondents were then resurveyed through four follow-ups in 1990, 1992, 1994, and 2000. On the questionnaire, students reported on a range of topics including: school, work, and home experiences; educational resources and support; the role in education of their parents and peers; neighborhood characteristics; educational and occupational aspirations; and other student perceptions. Additional topics included self-reports on smoking, alcohol and drug use and extracurricular activities. For the three in-school waves of data collection (when most were eighth-graders, sophomores, or seniors), achievement tests in reading, social studies, mathematics and science were administered in addition to the student questionnaire.

Example variables from NELS

Example variables from NELS—what level of measurement? Discrete or continuous?

- GENDER: 0 = male, 1 = female
- URBAN: 1 = urban, 2 = suburban, 3 = rural
- SCHTYP8: type of school attended, 8th grade 1 = public, 2 = private religious, 3 = private non-religious
- TCHERINT: agreement with "my teachers are interested in students," *Likert scale*
- NUMINST: number of post-secondary institutions the student attended

Example variables from NELS

Example variables from NELS—what level of measurement? Discrete or continuous?

- ACHRDG08: score on standardized test of reading achievement, 8th grade (ranges from 36.61 to 77.2)
- SES: socioeconomic status (ranges 0-35)
- SLFCNC12: self-concept score in 12th grade
- SCHATTRT: average daily attendance rate for the school student attended
- ABSENT12: number of times student missed school, categorical, 0 = never, 1 = 1-2 times 2= 3-6 times, etc.

Next time

Next time: Lecture 2

Describing univariate distributions:

- Frequency distributions
- Graphs for distribution of categorical variables
- Histograms
- Measures of central tendency and variability, skewness, quantiles
- Transforming variables